ACANS 7-Axis Servo Motion Controller Product Specification

1. Overview

The ACANS 7-Axis Servo Motion Controller is a professional-grade solution designed specifically for high-end motion simulators. It precisely drives up to 7 servo motors, providing ultimate dynamic feedback and attitude simulation for Sim Racing and Flight Simulation.

Key Advantages

- **Multi-Axis Linkage**: Native support for 7-axis output, compatible with 2DOF to 6DOF systems plus additional axes (e.g., seatbelt tensioners, surge, etc.).
- **Software Ecosystem**: Seamlessly compatible with industry-standard motion software including FlyPT Mover, SimTools, and SimHub.
- **Industrial Interface**: Utilizes RJ45 outputs for clean wiring, easily adaptable to various mainstream servo drives via standard adapter cables.

2. Hardware Connectivity

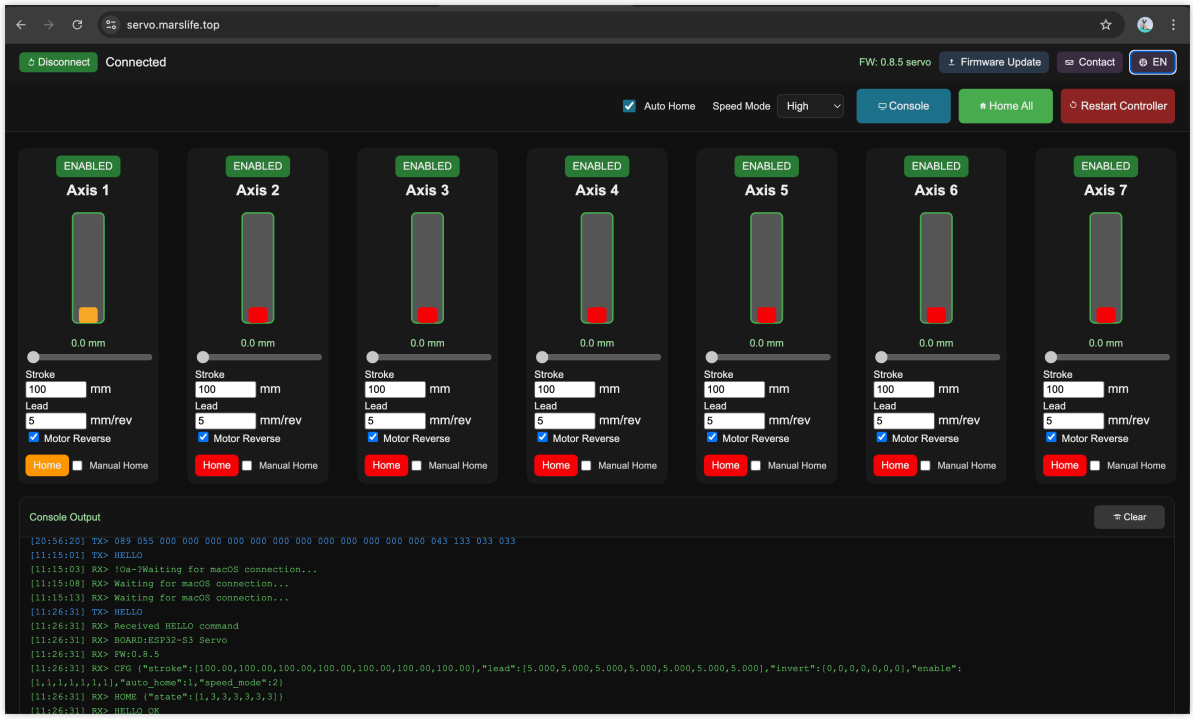
Wiring Definition (RJ45 to DB25)

The controller uses standard RJ45 ports for signal output, ensuring easy maintenance and reliable connections. Recommended RJ45 to DB25 (common servo interface) wiring definition:

Crystal Pin	Wire Color	Function	DB25 Pin	Color
3	White-Green	Pulse +	3	Brown
5	White-Blue	Direction +	4	Yellow
6	Green	Direction -	5	White
7	White-Brown	Enable	6	Gray
1	White-Orange	+5V	9	Red
8	Brown	0V (GND)	10	Black
4	Blue	Pulse -	14	Green
2	Orange	Torque Reached	23	Orange

3. Software Control Dashboard

The controller features an intuitive built-in web-based management interface, allowing for parameter configuration and status monitoring without any additional software installation.



Core Features

- **Axis Configuration**: Independent settings for Stroke, Lead, and Motor Direction for each axis.

- **Real-time Motion Control**: Supports slider-based debugging, manual homing, and one-click "Home All" functionality.
- **Performance Modes**: Selectable High, Medium, and Low speed modes to suit different simulation requirements.
- **System Diagnostics**: Real-time TX/RX raw data console for protocol analysis and troubleshooting.

Software Compatibility & Configuration

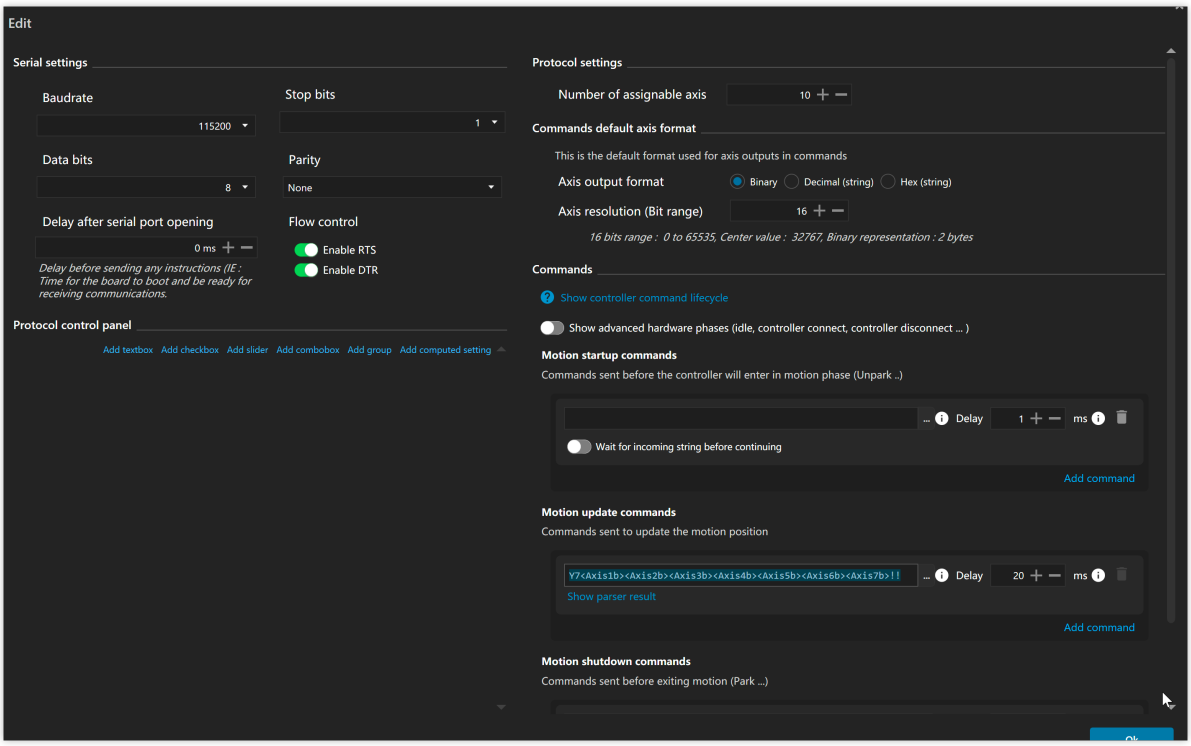
The controller natively supports various motion simulation software. Please follow the instructions below to configure the communication protocols:

1. FlyPT Mover Configuration

In the FlyPT Mover Serial output module, use the following connection string:
`flypt:Y7<Axis1a><Axis2a><Axis3a><Axis4a><Axis5a><Axis6a>00!!`

2. SimHub Configuration

In SimHub "ShakeIt For Controllers" -> "Custom Protocol", configure the serial port and command as shown below:



Key Configuration Points:

- **Baudrate**: 115200
- **DTR / RTS**: Both Enabled
- **Bit Range**: 16 bit
- **Command Content**: `Y7<Axis1b><Axis2b><Axis3b><Axis4b><Axis5b><Axis6b><Axis7b>!!`

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4. Quick Start Guide

1. **Power & Network**: Ensure the controller is powered and connected to your local network.
2. **Access Dashboard**: Open a browser and navigate to the dashboard (default: `servo.marslife.top` or the device IP).
3. **Mechanical Calibration**: Enter the Stroke and Lead parameters corresponding to your physical platform.
4. **Initial Homing**: Click "Home All" to calibrate mechanical zero points.
5. **Software Linkage**: Configure the output protocol in FlyPT Mover or SimTools to begin your simulation experience.

5. Technical Specifications

Parameter	Description
Number of Axes	7 Axes (Axis 1 - Axis 7)
Signal Type	Pulse + Direction
Output Interface	7 x RJ45
Communication	Compatible with FlyPT / SimTools / SimHub standards
Configuration	Cross-platform Web UI
Extras	Supports OTA Firmware Updates

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